YOGESH O K

yogesh.karmegam@gmail.com | Chennai, India | GitHub | LinkedIn

EDUCATION

Bachelor of Engineering, Computer Science and Engineering Expected 2028
Chennai Institute of Technology, Chennai **| CGPA: 7.98**
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, System Design

PROJECTS

RepoLens | FastAPI, React, PostgreSQL, pgvector, Redis, Celery, AST, RAG, Docker **|** GitHub
- Built a codebase-intelligence engine that ingests deterministic Git snapshots and extracts functions, classes, imports, calls, and exact line ranges using AST-based parsing.
- Combined dependency-graph traversal with embeddings and pgvector retrieval to answer architecture questions with source-file and line-level citations instead of ungrounded responses.
- Separated clone, parse, and embedding workloads into Celery workers via Redis with progress streaming and tenant-aware resource controls; defined validation across 20+ repositories, 10,000+ source files, and 100 evaluation queries.

Deploy-Lite | FastAPI, Docker, Celery, Redis, PostgreSQL, Nginx, MinIO, WebSockets, GitHub Webhooks **|** GitHub
- Architected a deployment engine that clones GitHub repositories and executes isolated Docker builds with configurable CPU, memory, and timeout controls before publishing frontend applications.
- Engineered asynchronous build processing with Celery/Redis, GitHub webhooks, PostgreSQL deployment history, and WebSocket log streaming so long-running builds remain independent of API request handling.
- Added MinIO object storage for built deployment artifacts and Nginx reverse-proxy routing to expose deployment-specific shareable URLs to users; designed validation for 10 concurrent builds and 50 success/failure/redeploy workflows.

QueueLess | FastAPI, React, PostgreSQL, Redis, WebSockets, QR Workflows, Docker **|** GitHub
- Built a real-time virtual queue platform with QR join/check-in, staff controls, cancellation and no-show handling, and transaction-safe position updates for service counters and campus events.
- Used Redis and WebSockets for live queue propagation and implemented rolling service-rate estimation to update expected wait times as throughput and queue conditions change.
- Structured load testing for 100+ simultaneous participants and 1,000 queue events, targeting sub-500 ms p95 state updates and under 20% wait-time prediction error before production use.

OpsFlow | FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Alembic, JWT, Redis, Docker **|** GitHub
- Built a multi-tenant incident and release management platform for organizations to manage users, teams, services, deployments, incidents, timelines, and postmortems.
- Implemented JWT authentication, RBAC, tenant-scoped authorization, relational schema design, Alembic migrations, and transaction-safe workflows across related resources.
- Added Redis-backed notifications, realtime WebSocket events, Docker Compose, and CI/CD automation; designed 100+ API, authorization, and integration test cases for tenant isolation, permissions, and rollback behavior.

INTERNSHIP EXPERIENCE

eGRCXperts - Software Development Intern | Chennai May 2026 - Jun 2026
- Built an RSA Archer workflow-automation system that transformed client workflow steps and Excel inputs into videos, mind maps, and UI visualizations for faster process understanding.
- Developed web-scraping and search-API pipelines with cosine-similarity classification to map cybersecurity-tool capabilities and support recommendation logic.
- Implemented API-oriented data transformation, automation, and structured-output workflows for internal GRC product analysis and engineering use cases.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript
Backend & Systems: FastAPI, REST APIs, WebSockets, Celery, JWT, RBAC, SQLAlchemy, Alembic, Redis
Frontend: React, React Router, HTML5, CSS3
Databases & Data: PostgreSQL, pgvector, MySQL, MinIO
DevOps & Tools: Docker, Docker Compose, Nginx, Git, GitHub Actions, CI/CD, Postman
AI / Data: RAG, Embeddings, Abstract Syntax Trees (AST), Dependency Graphs, Web Scraping, Prompt Engineering, Cosine Similarity
Software Engineering & Core CS: System Design, Distributed Systems, API Design, Database Design, Unit/Integration Testing, DSA, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS & LANGUAGES

- LeetCode Knight with 900+ problems solved across core data structures, algorithms, graphs, and dynamic programming.
- English: Professional working proficiency.
- Japanese: JLPT N3 Qualified; currently studying for JLPT N2.
- Tamil: Native proficiency.